This problem set covers material from Week 10, dates 4/20- 4/22. Unless otherwise noted, all problems are taken from the textbook. Problems can be found at the end of the corresponding subsection.

Instructions: Write or type complete solutions to the following problems and submit answers to the corresponding Canvas assignment. Your solutions should be neatly-written, show all work and computations, include figures or graphs where appropriate, and include some written explanation of your method or process (enough that I can understand your reasoning without having to guess or make assumptions)

Monday 4/20

1. Let's continue the example from class: Let $X_1, \ldots, X_n | \theta \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$. We'd like to use a LRT to test the following hypotheses:

$$H_0 : \theta = 0.5 \qquad\qquad H_1 : \theta \neq 0.5$$

   Suppose we have $n = 10$ data points and observe $\bar{X}_{obs} = 1$.

   (a) Using `R`, find the exact p-value for these data.

   (b) Obtain an approximate p-value for these data.

   (c) If you'd like your test to be level-0.05, what would be your decisions in (a) and (b)? What do you notice about your decisions, and why might that be the case?

2. Hardy-Weinberg revisited. Recall the two-allele scenario: an individual can have any of the two following genotypes: AA, Aa, or aa. If we randomly select n individuals from a population, we count up how many of the individuals have each allele. Thus we observe $(N_{AA}, N_{Aa}, N_{aa}) \sim \text{Multinom}_3(n, (p_{AA}, p_{Aa}, p_{aa}))$ where $p_j \geq 0$ and $\sum_{j=1}^{3} p_j = 1$. *In other words,* $\boldsymbol{p} \in \Omega =$ *"2D simplex".*

   Under the assumption of Hardy-Weinberg (H-W) equilibrium, allele A is present in the population with unknown probability θ, and allele a with probability $1 - \theta$, such that

$$p_{AA} = \theta^2 \qquad p_{Aa} = 2\theta(1-\theta) \qquad p_{aa} = (1-\theta)^2$$

   In practice, we do not know if H-W is present in a given population, but we might use data to test the following hypotheses:

   $H_0 : p_{AA} = \theta^2,\ p_{Aa} = 2\theta(1-\theta),\ p_{aa} = (1-\theta)^2$ (i.e. H-W is present)
   $H_1 : p_{AA}, p_{Aa}, p_{aa}$ are any three non-zero probabilities such that $p_{AA} + p_{Aa} + p_{aa} = 1$ (i.e. H-W not present)

   We will test these two hypotheses using a likelihood ratio test.

(a) Obtain the form of likelihood ratio statistic for these hypotheses. Simplify as much as possible.

(b) Recall the following data from Problem Set 2: In a sample from the Chinese population of Hong Kong in 1937, blood types occurred with the following frequencies, where A and a are erythrocyte antigens:

AA	Aa	aa	Total
342	500	187	1029

On that problem set, many of you said you believed H-W was present in this population based on some MLEs. Let's test this more rigorously: what is an approximate p-value of the observed data for this testing procedure?

(c) Based on your findings in the previous part, re-answer the following question: do you believe Hardy-Weinberg is present in this population? Briefly explain why or why not.

Wednesday 4/22

TBD

Friday 4/24

TBD

General rubric

Points	Criteria
5	The solution is correct *and* well-written. The author leaves no doubt as to why the solution is valid.
4.5	The solution is well-written, and is correct except for some minor arithmetic or calculation mistake.
4	The solution is technically correct, but author has omitted some key justification for why the solution is valid. Alternatively, the solution is well-written, but is missing a small, but essential component.
3	The solution is well-written, but either overlooks a significant component of the problem or makes a significant mistake. Alternatively, in a multi-part problem, a majority of the solutions are correct and well-written, but one part is missing or is significantly incorrect.
2	The solution is either correct but not adequately written, or it is adequately written but overlooks a significant component of the problem or makes a significant mistake.
1	The solution is rudimentary, but contains some relevant ideas. Alternatively, the solution briefly indicates the correct answer, but provides no further justification.
0	Either the solution is missing entirely, or the author makes no non-trivial progress toward a solution (i.e. just writes the statement of the problem and/or restates given information).
Notes:	For problems with multiple parts, the score represents a holistic review of the entire problem. Additionally, half-points may be used if the solution falls between two point values above.
Notes:	For problems with code, well-written means only having lines of code that are necessary to solving the problem, as well as presenting the solution for the reader to easily see. It might also be worth adding comments to your code.